Vitamins - high-density comparison study sheet

Foundations + fat-soluble vitamins (A, D, E, K)

Core idea + general characteristics

Objectives	Introduction; general characteristics; classification; individual characteristics; individual dietary sources; individual function; effect of deficiency.
Definition	The word vitamin comes from Latin vita = life. Vitamins are absorbed unchanged and have catalytic functions.
Need	Everybody must eat a small amount of vitamins (<1 gm/day) for growth and to stay healthy. After growth/development, vitamins remain essential for healthy maintenance of cells, tissues, and organs.
Energy context	Carbohydrates, fat, and protein are macronutrients. Digestion yields glucose/monosaccharides, fatty acids + glycerol, peptides + amino acids. Macronutrients are interchangeable energy sources: fat = 9 Kcal/gm; protein or carbohydrate = 4 Kcal/gm.
Storage	Water-soluble vitamins cannot be stored in human tissues; excess is excreted in urine. Significant amounts of fat-soluble vitamins can be stored in adipose tissue and liver.
Sources in body	Vitamins are obtained with food, but gut flora produce vitamin K and biotin. Vitamin D can be synthesized in skin with natural ultraviolet sunlight. Humans can form vitamin A from beta-carotene and niacin from tryptophan.
Other facts	Vitamins are required in small dietary quantities because they cannot be synthesized sufficiently by the body. Synthetic vitamins are identical to natural vitamins. They help the body efficiently use food energy and process proteins, carbohydrates, and fats for cellular respiration.

Classification by solubility

Criteria	Water-soluble	Fat-soluble
Toxicity	Rarely toxic	Mostly toxic
Transport	Free	Require carrier
Storage	Circulate freely	In cells with fat
Excretion	In urine	Stored with fat
Examples	B-complex vitamins + vitamin C	Vitamins A, D, E, K

Definitions	Fat-soluble vitamins dissolve in fat, can accumulate, and are saved for later use. Water-soluble vitamins dissolve in water, move easily through the watery body environment, and can be flushed out by kidneys.

Fat-soluble vitamins: horizontal comparison table

Comparison row	Vitamin A	Vitamin D	Vitamin E	Vitamin K / Phytonadione
Key identity / form	Group of unsaturated nutritional organic compounds: retinol, retinal, retinoic acid, and several provitamin A carotenoids; beta-carotene is most important. Active form is present only in animal tissue.	Group of fat-soluble vitamins found in liver and fish oils, or obtained by irradiating provitamin D with ultraviolet light. Enhances intestinal absorption of calcium, iron, magnesium, phosphate, zinc. Also called sunshine vitamin.	Yellow oily liquid freely soluble in fat solvent.	Structurally similar fat-soluble vitamins needed for complete synthesis of proteins required for blood coagulation (anti-hemorrhagic vitamin) and proteins involved in calcium binding in bone/other tissues. K1 in plants; K2 synthesized by intestinal bacteria; K3 can convert to K2 in the intestinal tract.
Principal sources	Liver; eggs; butter; dark green leafy vegetables.	Ultraviolet irradiation of skin is the major source; liver; fortified milk; eggs; butter. Diagram also shows diet/supplements, salmon, milk/orange juice.	Vegetable oil; leafy vegetables; legumes.	Vitamin K1: broccoli, cabbage, spinach, lettuce, turnip; intestinal flora after neonatal period.
Functions	Photoreceptor mechanism of retina; integrity of epithelium; lysosome stability; glycoprotein synthesis.	Calcium and phosphorus absorption; deposits phosphorus and calcium in teeth and bones, making them healthier and stronger; protects against infections by supporting immune system. Extra function list: calcium balance; cell differentiation; immunity; blood pressure regulation; development of bones & teeth.	Intracellular antioxidant; works with selenium; called anti-aging factor; protects cells from damage; keeps blood clean; aids body in using vitamin K; might help protect some people against Alzheimer's disease (AD); roles in neurological functions and inhibition of platelet aggregation.	Formation of prothrombin and other coagulation factors.
Deficiency effects	Night blindness; increased morbidity and mortality in young children.	Rickets; osteomalacia.	RBC hemolysis; neurological damage.	Tendency for bleeding; biliary dysfunction; impaired fat absorption.
Toxicity / dose / special notes	Toxicity: headache; peeling of skin. Usual therapeutic dose: 10,000-20,000 µg.	Toxicity: anorexia; renal failure; metastatic calcification. Activation pathway from diagram: skin previtamin D + UV vitamin D liver 25(OH)D kidney 1,25(OH)2D; downstream: calcium/muscle/bone health, BP regulation, insulin production, immune function, cell growth.	Toxicity: interferes with some enzymes.	Toxicity not specified in the lecture. High-yield anchor: coagulation factors/prothrombin + bleeding tendency.

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Water-soluble vitamins (B-complex + C)

Water-soluble vitamins: horizontal comparison table

Comparison row	B1 / Thiamine	B2 / Riboflavin	B3 / Niacin	B6 / Pyridoxine	B7 / Biotin	B12 / Cobalamin	Vitamin C / Ascorbic acid
Key identity / other names	Also called anti-Beri-Beri factor, anti-neuritic factor, and aneurin. Colorless basic organic compound; destroyed by alkaline and heat. Used by all living organisms but synthesized only in bacteria, fungi, and plants.	Also called beauty vitamin.	Known as niacin or nicotinic acid. Essential for metabolism of carbohydrate, protein, and fat.	Part of vitamin B complex; active form pyridoxal 5'-phosphate (PLP) serves as cofactor in many enzyme reactions in amino acid, glucose, and linoleic acid metabolism.	Water-soluble B vitamin. Normal hair and psychological function note included in lecture.	Complex organomatrix compound called cobalamin; cobalt-containing porphyrin; freely soluble in water.	Powerful antioxidant. Synthesized by most animals, not by humans. Destroyed by heat, e.g., heating to sterilize formula.
Principal sources	Meats; liver; potatoes; tuna; legumes.	Milk; meat; liver; cheese; eggs.	Meats; liver; fish; legumes.	Whole grain cereals; organ meat; liver; fish; milk; eggs; legumes.	Liver; kidney; cauliflower; eggs; nuts; legumes.	Mostly animal products: meat, shellfish, milk, cheese, eggs. Great sources: clams, oysters, mussels, caviar (fish eggs), crab, lobster.	Citrus fruit; tomatoes; cabbage; green peppers.
Functions	Carbohydrate metabolism; myocardial function; central and peripheral nerve function. Converts carbohydrate into energy; maintains healthy nervous system; necessary for healthy mucous membranes; helps digestion; provides muscle strength; useful for proper heart function. Used in treatment of beriberi.	Protein metabolism; integrity of mucous membranes; protects the body against cancer.	Carbohydrate metabolism; releases energy from carbohydrates, fats and proteins; essential for DNA synthesis; production of estrogen, progesterone and testosterone; may reduce migraine headaches; niacinamide may improve arthritis symptoms, joint mobility, and reduce anti-inflammatory medication need; necessary for healthy skin, nerves, digestive system; high-dose niacin medications prevent atherosclerosis development and reduce recurrent complications such as heart attack and peripheral vascular disease in affected people; detoxifies body; proper food digestion.	Porphyrin and heme synthesis; linoleic acid metabolism; tryptophan conversion to niacin; PLP cofactor roles in amino acid, glucose, and linoleic acid metabolism.	Amino acid and fatty acid metabolism; carboxylation and decarboxylation of oxaloacetic acid; maintenance of normal hair; normal psychological functions.	Essential for production of red blood cells; improves concentration, memory, and balance; important for metabolism of fat, carbohydrate, and proteins; promotes growth.	Synthesis of collagen. Maintenance of bones and proper functioning of adrenal and thyroid gland. Antioxidant. Stimulates immune function; combats bacterial infection; reduces effects of allergy-producing substances; protects vitamins A, E, and some B-complex vitamins from oxidation.
Secondary deficiency causes / risk	Pregnancy; lactation; fever; diarrhea; hyperthyroidism; impaired utilization in severe liver diseases.	Chronic alcoholism; diarrhea; liver diseases.	Diarrhea; cirrhosis; alcoholism; prolonged INH therapy.	Malabsorption or chemical inactivation by drugs such as hydralazine and penicillamine.	Raw egg white contains biotin antagonist avidin; high/prolonged consumption leads to dermatitis and glossitis.	Common in vegetarian diet; inadequate absorption in small intestinal disorder; inadequate utilization in liver and kidney malignancy; increased requirements in parasitic infestation; increased excretion in liver and renal disease.	Secondary deficiency very common in patients on ulcer diet.
Deficiency effects	Beriberi; cardiac failure; peripheral neuropathy; constipation. Beriberi symptoms listed: swelling, tingling/burning in hands and feet, confusion, difficulty breathing.	Angular stomatitis; corneal vascularization.	Pellagra; stomatitis; CNS and GI dysfunction.	Anemia; neuropathy; convulsions in infants.	Dermatitis; glossitis.	Anemia: megaloblastic or pernicious anemia. Demyelination and irreversible nerve cell death.	Scurvy: fragility of blood vessels; easy bruising/hemorrhage; poor healing; pain in bones and muscle; poor bone and dentin formation with loose teeth; compromised immunity.
Dose / special notes	Heat/alkali labile. High-yield: energy + nerves + heart + mucous membranes.	High-yield: mucous membranes + angular stomatitis + corneal vascularization.	High-yield: pellagra + energy release + DNA/hormones; high-dose drug use for atherosclerosis/recurrent vascular complications in those with condition.	High-yield: heme/porphyrin + tryptophan niacin + infant convulsions.	Avidin in raw egg white is the key special note.	High-yield: animal sources + vegetarians + RBC + nervous system + irreversible nerve cell death.	Usual therapeutic dose 100-1000 mg/day. Huge dose 10 gm/day does not decrease incidence or severity of common cold but may predispose to oxalate urinary calculus.

Fast recall: deficiency anchors and exam flags

A - Night blindness; morbidity/mortality in young children D - Rickets + osteomalacia; toxicity: anorexia, renal failure, metastatic calcification E - RBC hemolysis + neurological damage	K - Bleeding tendency; prothrombin/coagulation factors B1 - Beriberi; cardiac failure; peripheral neuropathy B2 - Angular stomatitis; corneal vascularization	B3 - Pellagra; stomatitis; CNS + GI dysfunction B6 - Anemia; neuropathy; infant convulsions B7 - Dermatitis + glossitis; raw egg white avidin	B12 - Megaloblastic/pernicious anemia; demyelination + irreversible nerve death C - Scurvy: hemorrhage/bruising, poor healing, loose teeth, bone/muscle pain